

Minh Hieu Luong

Full-stack Software Engineer

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PROFESSIONAL SUMMARY

- Full-stack Software Engineer with 6 years delivering and modernizing enterprise applications in .NET, Python and TypeScript, now focused on Azure cloud engineering.
- Hands-on with Azure Kubernetes Service, Functions, Data Factory and CI/CD pipelines; migrated production workloads from AWS to Azure, improving performance while reducing operating cost.
- Owns features end to end — requirements, system design, implementation, testing, deployment and production support — and designs for auditability and safe failure.
- Works directly with US and Japanese clients in English, and applies AI and machine learning where it measurably improves the product.

TECHNICAL SKILLS

Programming Languages: C#, JavaScript, TypeScript, Python

Frameworks & Engines: .NET Core / Framework, Entity Framework, React, Next.js, Flask / FastAPI, Docker

Cloud & DevOps: Azure — Kubernetes Service (AKS), Functions, Data Factory, DevOps; CI/CD pipelines; AWS to Azure migration

Databases: SQL Server, PostgreSQL, MySQL, Redis

Libraries: jQuery, Bootstrap, DevExtreme, Redux, NgRx, React Flow, NumPy

AI & Machine Learning: Applied ML — logistic regression, feature weighting, model evaluation (AUC / precision / recall); GitHub Copilot

Version Control: GitHub, GitLab, Bitbucket, Azure Repos

Working Knowledge: Angular, Vue.js, Xamarin

Methodology: Agile / Scrum, Kanban

PROFESSIONAL EXPERIENCE

Software Engineer | Hypertek | Apr 2026 – Present

Project: IPMS — Intellectual Property Management System | Team size: 8

An enterprise platform covering the full IP lifecycle — trademarks, patents, industrial designs and geographical indications — with registration and approval workflows, document management, fee calculation, domain-name management and automated crawling of external registries such as WIPO.

- Designed and shipped an AI-driven IP infringement early-warning system: a five-signal weighted risk model scoring each filing High/Medium/Low with per-factor bilingual explanations the legal team can audit, verified end to end by 46 automated tests on a production-shaped database.
- Added a self-learning layer — logistic regression in NumPy, retrained weekly by a Kubernetes CronJob — predicting infringement probability over a six-month horizon, with time-ordered holdout validation and AUC/precision activation gates.
- Kept the feature fail-safe: PostgreSQL model registry with append-only audit log, and in-process inference that falls back to the rule engine on any error; risk preview measured about 8 ms per request against a 300 ms budget.
- Chose trigram similarity (pg_trgm + unaccent) over embeddings to catch Vietnamese look-alike brand names, and materialized views over JSONB case data to keep trend and hotspot queries fast.
- Delivered core IP features end to end across API, search, admin and crawler services — alert management, bulk IP import, domain-name management, and React Flow visualization of IP relationship and approval-workflow graphs.
- Hardened the Python crawler (owner extraction, WIPO ID fallback matching, retry logic) and established centralized logging and observability on AKS.

Tools & Technologies: Python, FastAPI, NumPy, TypeScript, React, Next.js, React Flow, PostgreSQL (pg_trgm, materialized views), Docker, Azure Kubernetes Service (AKS), GitLab CI, Jira

Software Engineer | FPT Software | Mar 2026 – Jun 2026

Project: Legacy Insurance System Modernization — Japan (NDA) | Team size: 180

A legacy-system modernization project for Japan's agricultural and livestock insurance business, migrating business applications from Microsoft Access to a centralized .NET-based monolithic architecture to improve maintainability, scalability, performance and long-term sustainability.

- Analyzed legacy Microsoft Access applications and migrated their business logic onto the new .NET platform.
- Developed and maintained .NET applications within the target monolithic architecture.
- Migrated and optimized SQL databases to improve query performance on the consolidated schema.
- Resolved defects and delivered enhancements driven by customer feedback.

- Clarified requirements and handled customer comments directly with the onshore team.
- Conducted code reviews and supported deployment activities across releases.

Tools & Technologies: C#, .NET 10 MVC, SQL Server (SSMS), Visual Studio 2022, GitLab, TortoiseSVN

Software Engineer & Data Engineer | FPT Software | Sep 2025 – Feb 2026

Project: Healthcare Patient-Outreach Platform — U.S. (NDA) | Team size: 10

A healthcare platform that automates patient outreach, manages care data, and streamlines operational workflows.

- Implemented and maintained data-processing workflows and pipelines in Azure Data Factory.
- Migrated Azure Functions from the in-process model to the isolated worker model, unblocking future .NET upgrades.
- Delivered campaign scheduling, data synchronization and system enhancement initiatives.
- Improved SonarQube code coverage and drove continuous improvement in performance, reliability and platform modernization.

Tools & Technologies: C#, TypeScript, .NET Core, Angular, Azure Data Factory, Azure Functions, Azure Cloud Services, SSMS, Visual Studio 2022, Azure Repos

Software Engineer | FPT Software | Mar 2022 – Sep 2025

Project: US Real-Estate Document Notarization (NDA) | Team size: 47

A platform for the notarization of US real-estate documents, handling high-volume transactions.

- Worked directly with US clients to clarify requirements and resolve critical issues affecting high-volume transactions.
- Proposed and implemented technical solutions that improved system reliability and scalability.
- Optimized SQL performance through new and refactored stored procedures, significantly reducing query execution time.
- Migrated workloads from AWS to Azure, improving performance while reducing operational cost.
- Built and maintained Azure CI/CD pipelines, automating build and deployment to eliminate manual release errors.
- Developed RESTful APIs, web services and responsive web views, integrating multiple third-party services.
- Achieved 80% SonarQube code coverage and supported production releases with minimal downtime.
- Contributed to a Vue.js component-based architecture with reusable components adopted across the product.

Tools & Technologies: C#, JavaScript, .NET Framework MVC, .NET 8 API, Vue.js, DevExtreme, Azure Cloud, Visual Studio 2022, GitLab

Software Engineer | FPT Software | Apr 2021 – Jan 2022

Project: B2B Microservices Data & Operations Application (NDA) | Team size: 13

A microservices application offered to third parties, providing tools to collect company data and manage internal business operations.

- Gathered and analyzed business requirements with the BrSE, translating them into database design and new web services on a microservices architecture.
- Produced project documentation for handover and maintenance.
- Developed unit tests sustaining at least 80% code coverage.
- Troubleshoot and resolved production issues.

Tools & Technologies: Python, Flask, RESTful API, Microservices, Docker, PayPal REST APIs, SSMS, Visual Studio 2019, GitLab

Software Engineer | Ugotechs | Jan 2021 – Mar 2021

Project: Company Branch & Organization Management (Internal) | Team size: 3

- Gathered and analyzed business requirements with the client and translated them into the database design.
- Built new web pages for branch and organization management.
- Troubleshoot and resolved production issues.

Tools & Technologies: C#, .NET Framework MVC, SQL Server (SSMS), Visual Studio 2019, GitLab, Windows Server

Software Engineer | Ugotechs | Jun 2020 – Jan 2021

Project: COVID-19 Notification & Information App (NDA) | Team size: 10

- Gathered and analyzed business requirements with the client and translated them into function and screen designs.
- Developed unit tests sustaining at least 80% code coverage.
- Troubleshoot and resolved production issues.

Tools & Technologies: C#, Xamarin.Android, .NET Framework 4.8, SSMS, Visual Studio 2019, GitLab

EDUCATION

Bachelor of Software Engineering | Duy Tan University | 2020

LANGUAGES

Vietnamese: Native

English: Professional working proficiency — effective client communication